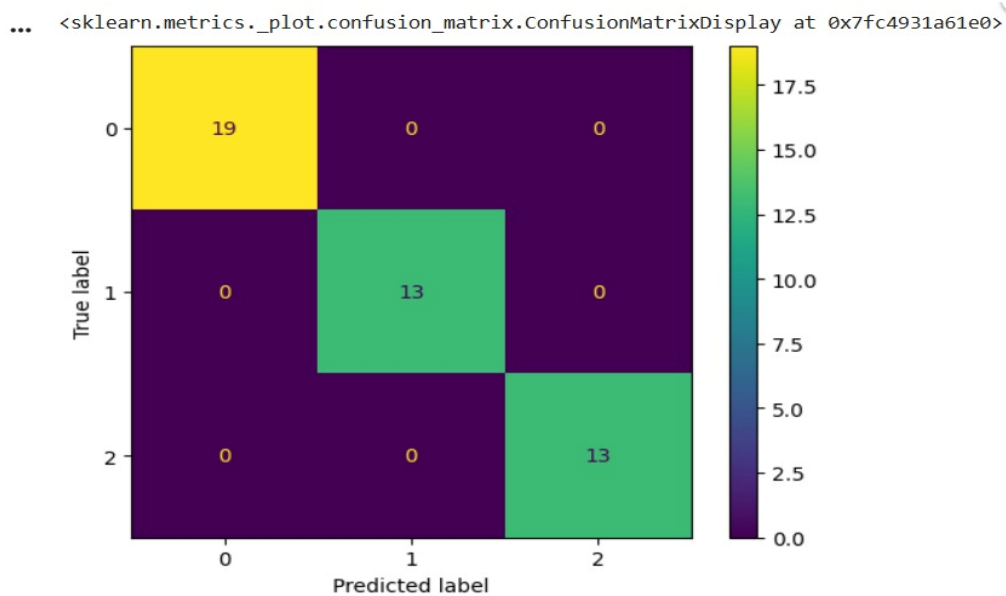


EXP:1

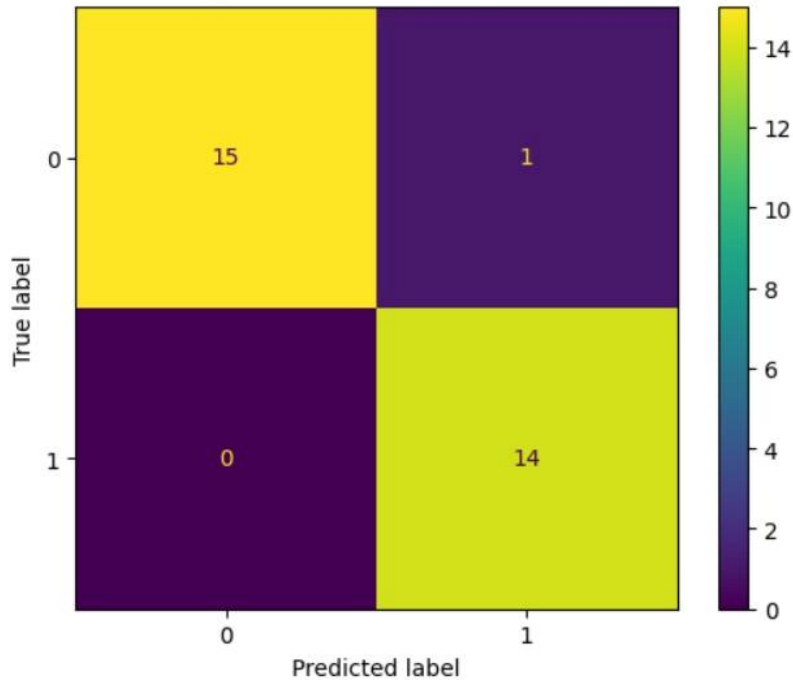
```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay
X, y = load_iris(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
model = LogisticRegression(max_iter=200).fit(X_train, y_train)
ConfusionMatrixDisplay(confusion_matrix(y_test, model.predict(X_test))).plot()
```



EXP:2

```
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import ConfusionMatrixDisplay
X, y = make_classification(n_samples=100, n_features=2, n_redundant=0, random_state=42)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
model = LogisticRegression().fit(X_train, y_train)
ConfusionMatrixDisplay.from_predictions(y_test, model.predict(X_test))
```

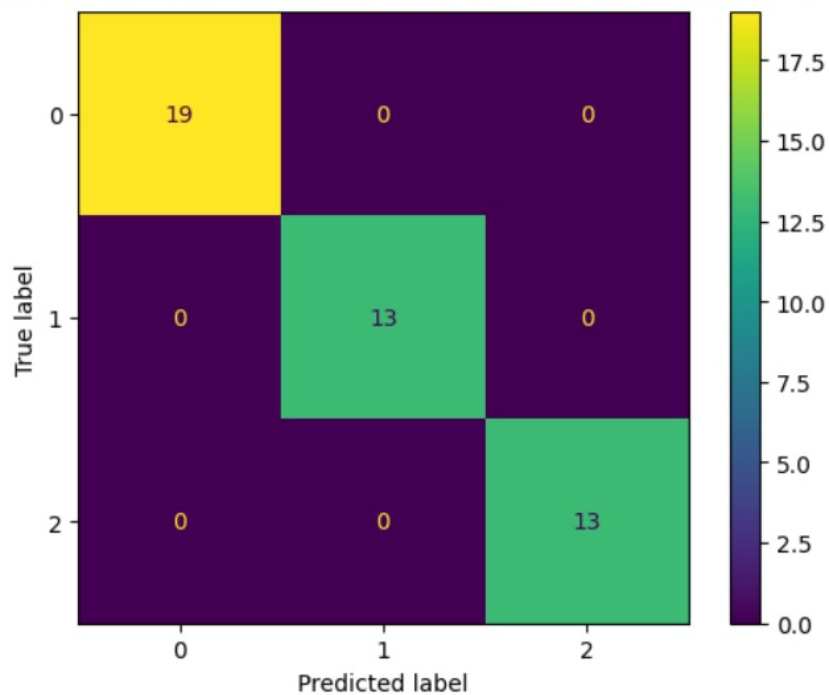
... <sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x7fc4af744320>



EXP:3

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import ConfusionMatrixDisplay
X, y = load_iris(return_X_y=True)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
model = DecisionTreeClassifier().fit(X_train, y_train)
ConfusionMatrixDisplay.from_predictions(y_test, model.predict(X_test))
```

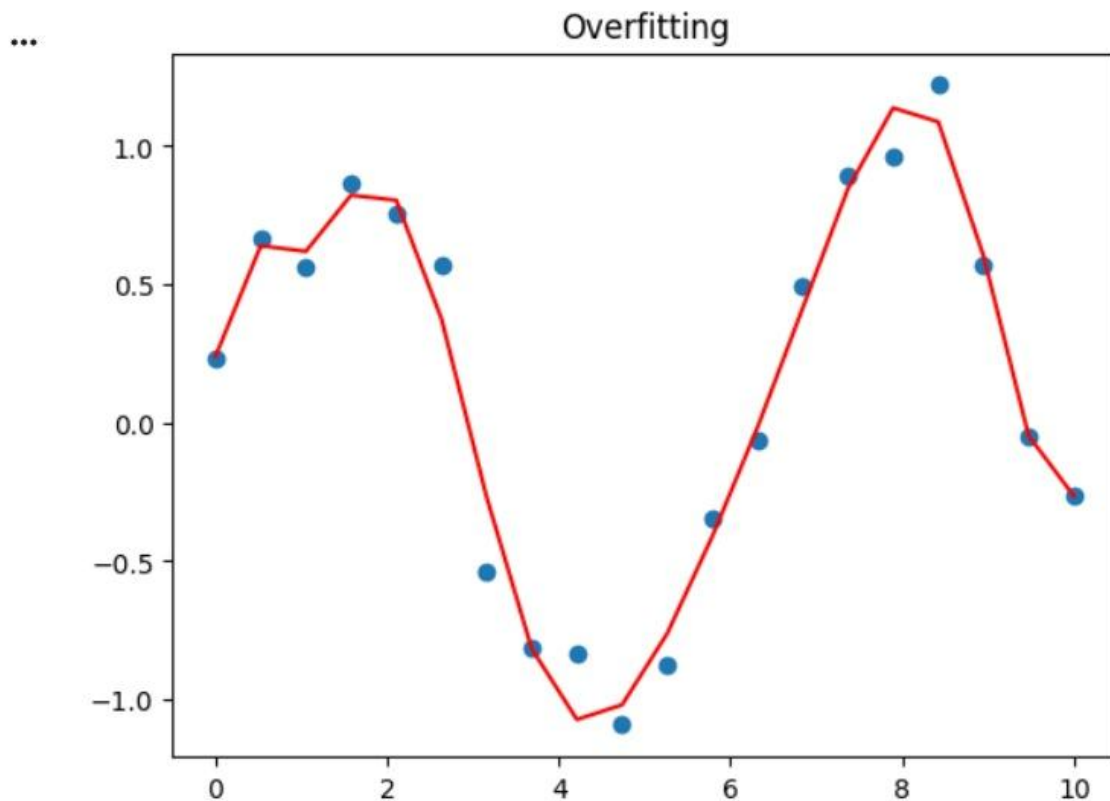
... <sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x7fc4924551f0>



EXP:4

```
import numpy as np, matplotlib.pyplot as plt
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression

x=np.linspace(0,10,20).reshape(-1,1); y=np.sin(x).ravel()+0.2*np.random.randn(20)
X=PolynomialFeatures(10).fit_transform(x)
m=LinearRegression().fit(X,y)
plt.scatter(x,y); plt.plot(x,m.predict(X),'r'); plt.title("Overfitting"); plt.show()
```



EXP:5

```
from sklearn.datasets import make_regression
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score

X,y=make_regression(n_samples=100,n_features=1,noise=10,random_state=42)
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.3,random_state=42)
model=LinearRegression().fit(X_train,y_train)
print("R2 Score:",r2_score(y_test,model.predict(X_test)))
```

... R² Score: 0.9408983055235901

